

Activity 2: UML Diagram Design Project

Title: A Web-Based Laundromat Management System (*Washmate*)

Brief system description:

The washmate system is a web-based laundromat management platform designed to automate and streamline laundry operations. It aims to address inefficiencies in the traditional manual methods used in laundromats, such as inaccurate records, misplaced laundry, long waiting times, and challenges in tracking sales and operations. The system enables the automation of laundry order management, real-time tracking of laundry status, and payment processing, improving service efficiency and accuracy in daily operations. By centralizing operations, the system allows for seamless management of customer orders, statuses, and transactions, while also simplifying report generation and notification delivery.

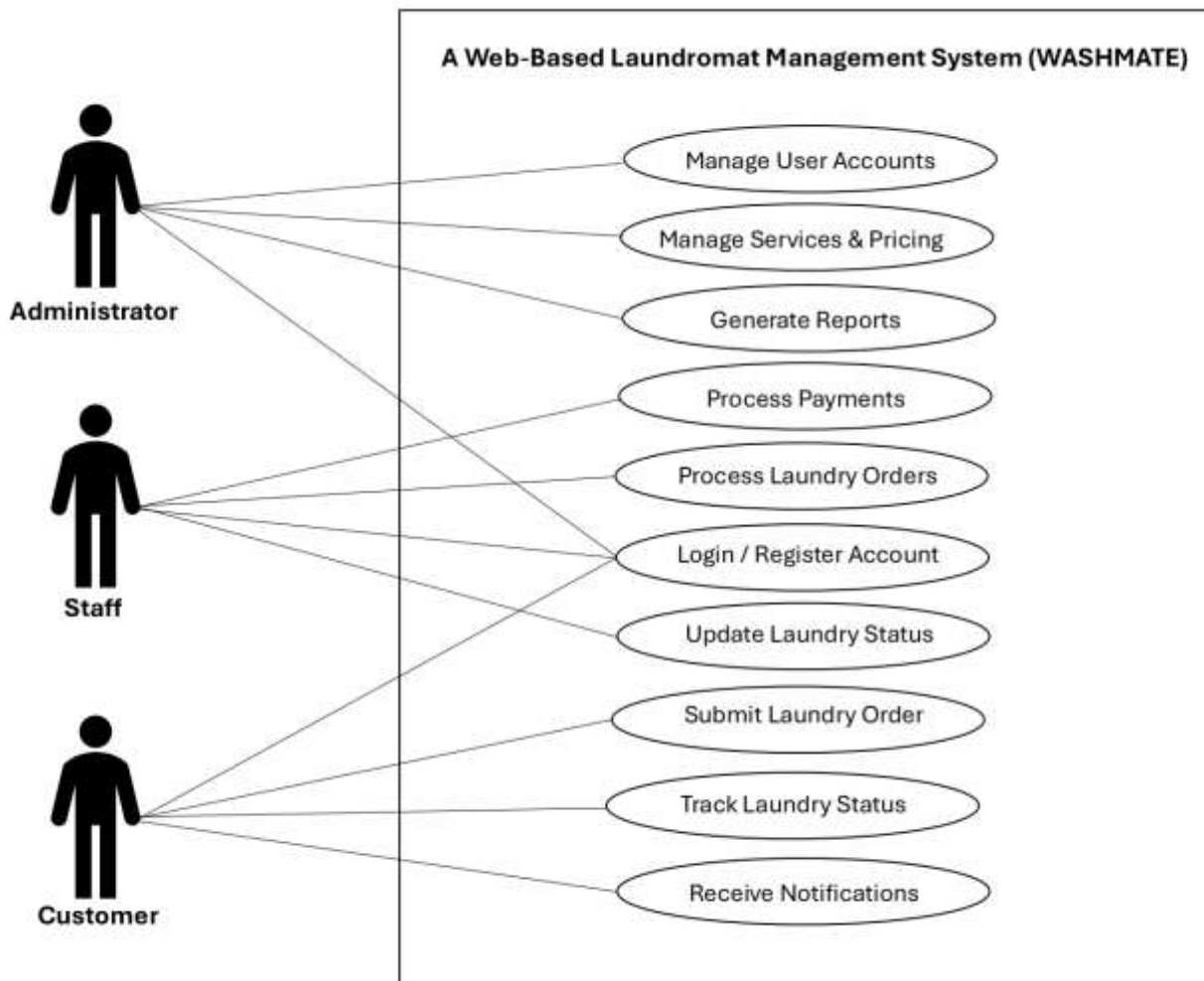
Identify actors/users:

1. Administrator – Responsible for managing user accounts, setting up services and pricing, and generating reports.
2. Staff – Handles customer orders, updates laundry status, processes payments, and interacts with customers to ensure smooth operations.
3. Customer – Submits laundry requests, tracks order status, and makes payments.

Identify main system functions:

1. User Authentication – Allows administrators, staff, and customers to create accounts and log in securely.
2. Order Management – Enables customers to place laundry orders, which are then managed by staff members.
3. Laundry Status Tracking – Customers and staff can monitor the status of laundry orders in real-time (e.g., received, washing, ready for pickup).
4. Payment Processing – Facilitates payment processing for laundry services, streamlining financial transactions.
5. Report Generation – Administrators can generate reports related to transactions, laundry orders, and sales for better management oversight.
6. Notifications – Sends alerts to customers about their order status, enhancing communication and customer satisfaction.

UML Diagram: Use Case Diagram



Short Explanation:

The Use Case Diagram for the laundromat management system (*washmate*) illustrates the interactions between the three primary users—Administrator, Staff, and Customer—and the core functionalities of the system. The Administrator is responsible for managing user accounts, services and pricing, generating reports, processing payments, and overseeing laundry orders. The Staff handles tasks such as processing laundry orders, updating laundry status, and managing payments. Meanwhile, the Customer submits laundry orders, tracks their laundry status, and receives notifications about the progress of their orders. This diagram clearly shows the roles and responsibilities of each user within the system.